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B. E. (Third Semester) Examination, April-May 2016

(New Scheme)

(Elect. Engg. Branch)

ELECTRICAL POWER GENERATION

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d) of each question. If needed, assume suitable data numerical question.*

1. (a) How can you improve the load factor of any power plant? Describe in points. 2

(b) The load curve of an electrical system has the following values : 7

Time	Load (MW)
12 mn	50

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5 am	50
9 am	100
6 pm	100
8 pm	150
10 pm	80
12 mn	50

(i) Plot chronological load curve.

(ii) Find the energy required by the system in one day.

(iii) Find system daily load factor.

(c) Explain how does load factor and diversity factor influence the size of generating units and cost of generated energy. 7

(d) What is the difference between : 7

(i) Base load and Peak load

(ii) Capacity factor and Utilization factor

2. (a) Discuss why overall efficiency of Thermal Power Plant is very low? 2

(b) Draw and explain : 7

(i) Flue gas flow diagram

(ii) Various stages in coal handling

(c) Describe the use of different parts of a thermal power plant with the help of neat sketch. 7

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(d) What are the factors to be considered for selection of site for a thermal power plant? Draw a typical layout and explain the working of a plant. 7

3. (a) Define the following terms : 2

(i) Mass defect

(ii) Multiplication factor

(b) Write short notes on the following : 7

(i) Hydraulic dams

(ii) Pumped storage hydel plant

(iii) Nuclear power plant as base load plant

(c) The mean monthly discharge at a particular site is given below : 7

Month	Discharge	Month	Discharge
Jan.	200	July	1600
Feb.	400	Aug.	1200
March	600	Sept.	2000
April	2400	Oct.	1200
May	1200	Nov.	800
June	1800	Dec.	400

Draw the hydrograph. Determine the average in flow and power that can be developed at an effective head of 90 m. Determine the capacity of the storage reservoir based on above one year data (neglecting the losses due to sepage, evaporation etc.). Assume overall generation efficiency is 80%.

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(d) What are the factors to be considered for site selection of neclear power plant and hydro power plant. 7

4. (a) What are the conventional and non-conventional energy resources. 23 2

(b) What are the advantages and disadvantages of concentrating collectors over flat plate collectors? 7

(c) Describe the basic components of wind energy conversion system with neat black diagram. 7

(d) Discuss the advantages and disadvantages of horizontal and vertical axis wind mill. What methods are being used to overcome the fluctuating power generation of wind mill? 7

5. (a) How can geothermal energy utilised for electricity generation? 2

(b) Describe a closed cycle MHD system with advantage and disadvantage of MHD system. 7

(c) What is ocean thermal energy? How the ocean thermal energy plants are classified? Explain open cycle OTEC plant. 7

(d) Draw the block diagram of biomass energy unit. Explain each in detail. Compare the merits and demerits of electrical energy with other forms of energy. 7

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